Module 2 Assignment: Exploring Generative AI Applications

Option 3: ChatGPT and Gemini

Patrick Hall — AAI101 Introduction to AI

Overview

For this activity I used ChatGPT and Google Gemini with the same prompts. I asked each tool to explain how it works, create a presentation about the benefits of cloud computing, write a research paper about the future of technology, and design an app that solves a real-world problem. Then I compared the two tools using the seven required parameters, noted what I would verify before school or work use, and identified one limitation I observed.

Prompt 1: Write a paragraph on how you work

ChatGPT's Response:

"ChatGPT is an artificial intelligence system designed to understand and respond to human language. When you send me a question or message, I analyze the words and their context to figure out what you are asking. I then generate a response by predicting which words and ideas are most appropriate based on patterns learned during training. I can explain topics, answer questions, help with writing, solve problems, and assist with many other tasks. I do not think or feel like a human, but I can process information and communicate in a way that is designed to be helpful and natural. For some questions, I can also use tools to find current information or perform specific tasks."

Gemini's Response:

"I operate as a large language model trained by Google, processing and generating text by analyzing statistical patterns in human language. When you send a prompt, I break down your text into smaller tokens, evaluate their context using transformer-based neural network architectures, and predict the most logical, accurate, and helpful response sequence word by word. To make these outputs reliable and safe, my training combines pre-training on extensive datasets with fine-tuning techniques like Reinforcement Learning from Human Feedback (RLHF). Additionally, I can connect to external tools—such as web search APIs for real-time information or specialized capabilities for data analysis—allowing me to synthesize complex requests, follow detailed instructions, and collaborate dynamically in real time."

Summary: ChatGPT explained itself in everyday language. Gemini used more technical terms such as tokens, transformers, and RLHF. Both said they predict responses from patterns in training data and can use tools for some tasks.

Prompt 2: Create a presentation about the benefits of cloud computing

ChatGPT produced a beginner-friendly PowerPoint titled “Benefits of Cloud Computing.” The slides explained what cloud computing is and covered lower costs, scalability and flexibility, anywhere access and collaboration, backup and recovery, security and automatic updates, and faster innovation. The tone was simple and easy to follow for someone new to the topic.

Gemini produced a more advanced PowerPoint titled “The Strategic Power of Cloud Computing.” It used business language and covered CapEx to OpEx cost shifts, elasticity, security, AI and analytics, high availability, disaster recovery, IaaS/PaaS/SaaS models, and a cloud migration roadmap. It also included specific-sounding metrics, such as faster deployment and cost-reduction percentages, that would need fact-checking.

Overall, ChatGPT was clearer for a short student presentation. Gemini looked more like an enterprise briefing and went deeper into service models and migration planning.

Prompt 3: Write a research paper about the future of technology

ChatGPT’s paper, “The Future of Technology,” used clear sections on artificial intelligence, robotics and automation, biotechnology and healthcare, energy and climate technology, immersive digital environments, cybersecurity and privacy, and social impact. It balanced opportunities with risks and was readable as a student research paper.

Gemini’s paper, “The Next Horizon: Emerging Technological Paradigms and the Future of Society,” focused on convergence across AGI, quantum computing, biotechnology and neurotechnology, and sustainable energy. It included diagrams and denser academic language. It sounded more advanced, but some claims and framing would need careful verification before academic use.

Both tools handled long-form writing well. ChatGPT was clearer and more approachable. Gemini was more technical and framework-heavy.

Prompt 4: Design an app that solves a real-world problem

ChatGPT designed FoodSave, an app to reduce household food waste. Features included barcode scanning, expiration reminders, smart recipe suggestions, food inventory, shopping lists, and a waste tracker. The explanation was straightforward and easy to understand.

Gemini designed FreshCycle, also focused on food waste. Features included OCR receipt scanning, dynamic expiration tracking, pantry-first recipes, a community sharing network, and a waste/savings dashboard. Gemini also added a tech stack, workflow diagram, and monetization plan (freemium and premium).

Both apps solved a similar problem, but Gemini went further into technical architecture and business details, while ChatGPT stayed focused on user-facing features.

Comparison of ChatGPT versus Gemini

Parameter 1: Human-Like Response

ChatGPT generally sounded more conversational and easy to follow, like a tutor explaining ideas in everyday language. Gemini also wrote clearly, but its tone was more formal and technical, especially in the cloud presentation and research paper. For a student audience, ChatGPT felt more natural, while Gemini felt more like a professional or executive-style response.

Parameter 2: Training Dataset and Underlying Technology

Both tools are large language models that generate answers from patterns learned during training. ChatGPT described itself simply as analyzing context and predicting helpful words and ideas. Gemini explained its process with more technical detail, mentioning tokens, transformer-based neural networks, pre-training, and Reinforcement Learning from Human Feedback (RLHF). That difference shows up in the writing: ChatGPT stays accessible, while Gemini leans into technical explanation.

Parameter 3: Authenticity of Response

Both tools produced polished, confident responses that looked ready for school use. ChatGPT's cloud slides and technology paper stayed fairly general and readable, which made them feel trustworthy but still required checking. Gemini sounded especially authoritative and included specific claims, metrics, and advanced framing that may or may not be fully accurate. Authenticity still depends on verifying sources rather than trusting how finished the text looks.

Parameter 4: Access to the Internet

In the "how you work" responses, both tools said they can use additional tools for current information. ChatGPT noted it can use tools to find current information or perform specific tasks. Gemini said it can connect to external tools such as web search APIs for real-time information. For this assignment, that means either tool might include newer information, but important facts should still be confirmed independently.

Parameter 5: User Friendliness and Interface

Both ChatGPT and Gemini are chat-based and easy to open for this project. ChatGPT's outputs were simpler to turn into a beginner presentation and a clear paper. Gemini's interface also worked well and produced denser formatting, including more complex slide structure and diagrams. For this class activity, ChatGPT felt a bit easier to work with, while Gemini was still usable but more advanced in presentation style.

Parameter 6: Text Processing: Summarization, Paragraph Writing, etc.

Both tools handled long writing and structured output well. ChatGPT was strong at clear outlines, sectioned essays, and straightforward slide points. Gemini was strong at frameworks, tables, workflow diagrams, tech stacks, and more elaborate organization. ChatGPT processed prompts into student-friendly text, while Gemini processed them into more detailed professional-style documents.

Parameter 7: Charges and Price

Both tools offer free access with optional paid upgrades. ChatGPT has a free tier and paid plans for higher limits and features. Gemini is available with a Google account and may also include paid or advanced options depending on the plan. For this assignment, both were usable without paying, but heavier use or advanced features could require a subscription.

What information would I verify before using this response in a school or workplace setting?

Before using either tool's output in school or at work, I would verify factual claims, statistics, and citations. For example, Gemini's cloud presentation included specific percentages about speed and cost savings that I would check against reliable sources before repeating them. I would also verify any research-paper references and rewrite AI-generated text in my own words so the final work reflects my understanding. For the app designs, I would treat feature lists and technical stacks as ideas, not as proven product plans, until researched further.

Limitation, weakness, bias, incorrect result, or surprising outcome

One limitation I observed is that both tools can sound finished and authoritative even when details may be incomplete, overstated, or hard to verify. This was especially noticeable in Gemini's cloud presentation, which included specific metrics that looked precise but were not clearly sourced. This happens because generative AI predicts likely wording from patterns in training data; it is not the same as checking primary evidence for every claim. A user should verify important numbers and citations by locating original sources, comparing multiple reputable references, and removing any claim that cannot be confirmed before relying on the output in school or workplace settings.

Closing reflection

Overall, ChatGPT and Gemini are both useful for brainstorming, outlining, and drafting. ChatGPT was stronger for clear, student-friendly explanations. Gemini was stronger for detailed structure and technical depth. One place the tools did well was turning a short prompt into organized slides, papers, and app concepts quickly. One place their output should be questioned is anywhere they present specific statistics or advanced claims without sources. For my path in data science and AI, these tools can help me explore

ideas and practice explaining concepts, then I can verify facts with primary sources. What I disliked was how easy it is to over-trust a clean-looking answer.

Attachments / supporting files

Supporting files from this activity include the ChatGPT and Gemini cloud-computing PowerPoint presentations and the ChatGPT and Gemini future-of-technology research papers used for Prompts 2 and 3. Upload those files with this document in Canvas.